

Multi-Agent Workforce Intelligence System for the Energy Industry

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1. What the agent does

I work as a People Strategy Lead for ABB's Energy Industries division in Houston. The midterm blueprint for this project came directly from my day-to-day work: workforce planning, HR business partnering, and oversight of staffing across seven countries in the Americas. The system I built analyzes resumes and job descriptions, reviews twelve months of HRIS time-series data for a department, and produces a boardroom-ready memo that combines a hiring decision, a retention plan for the receiving team, and a backfill pipeline for the role. Every number in the memo directly references a verified deep learning model output, not a synthesized approximation. The memo cites the rule that produced the verdict and includes a full execution trace, so a senior reviewer can see which tools were called and what they returned.

The problem is real, and I have lived through it. The IEA's World Energy Employment 2025 report shows 76 million energy jobs worldwide, with 2.4 workers nearing retirement for every new hire under 25 in advanced economies. CERAWEEK 2026 panels reported that 40 percent of plant operators could retire by the end of this decade. Accenture's research, presented at the same event, revealed that just 8 percent of energy organizations have a clear understanding of their workforce's existing skills. Meanwhile, 85 percent of executives believe AI will transform their business, but only 18 percent observe tangible benefits. The system I built directly addresses the visibility gap. It knows a candidate's skills and the skills the team is about to lose, and it produces a hiring recommendation that integrates both. The target user is a People Strategy Lead, an HR Business Partner, or a CHRO at a large energy company.

2. Option Choice and Changes

I chose Option B (multi-agent system) and stayed with it. Two distinct agent roles, Talent Intelligence and Workforce Forecasting, communicate through a single orchestrator. Agent 1 reads resumes and job descriptions, runs a routed NER ensemble plus a Sentence BERT semantic match, and produces a 19-signal skill gap analysis that identifies the specific missing skills and certifications rather than a generic match score. Agent 2 processes HRIS time series data, utilizes a Bidirectional LSTM with attention to forecast attrition risk and headcount changes, and generates a 19-signal workforce risk analysis. This analysis segments cohorts, links predictions to key drivers, simulates the

possible effects of retention strategies, and quantifies replacement costs exposure. The framework choice changed from the midterm blueprint. I committed to CrewAI plus a hosted LLM as the agent's brain. I built that path (it works, lives in agents.py, and remains active when an LLM API key is present), but the primary deliverable is a deterministic, LLM-free orchestrator I wrote from scratch in brain.py. The reason is the same as why I take this project seriously at work. The workforce planning decisions are audited, regulated, and contestable. Every "do not advance" recommendation can be challenged on grounds of bias, and the EU AI Act classifies HR systems as high risk for a reason. An LLM brain produces fluent narratives that cannot explain in plain language which rule produced a specific verdict. A 20-cell decision matrix where every cell contains a named rule (R1 strong stable through R20 decline critical) and an explicit rationale. CrewAI remains wired as the optional path for cases where free text generation actually adds value (resume coaching, drafting job descriptions, multi-turn interview prep). For the actual hiring and retention decisions this project addresses, the LLM adds latency and cost without adding capability. Under the same joint hire scenario, the proprietary brain returns a memo in about 500 milliseconds. The CrewAI path with a hosted LLM takes 15 to 30 seconds and costs roughly \$0.03 to \$0.10 per memo.

3. Architecture & Deep learning models

The architecture follows the INPUT to DL MODEL to AGENT BRAIN to TOOL to OUTPUT flow specified in the midterm blueprint, with one important refinement to the agent brain box. The agent brain is now a deterministic orchestrator by default, with the LLM path retained as an option. Each agent has its own deep learning models and tools, and both feed into the same brain.

Model	Agent	Course module	Role
DistilBERT cased, fine tuned	1	Module 05 Transformers	Resume NER on the legacy five label classes
GLiNER zero shot (NAACL 2024)	1	Module 05 Transformers	Generalist NER coverage across all ten labels
ModernBERT v11	1	Module 05 Transformers	NER specialist that owns the TOOL class

Sentence BERT all MiniLM L6 v2	1	Module 05 Transformers	Cosine semantic match plus Hungarian assignment for the skill gap diff
Bidirectional LSTM with attention	2	Module 03 RNN and LSTM	Department level attrition probability and headcount forecast
Routed per class ensemble	1	Module 05 Transformers	Per class hard ownership across the four NER variants above
Custom proprietary brain	both	Module 10 Agentic AI	Plan, Act, Observe, Respond loop over 21 tools, no LLM dependency

The four NER models cooperate via a per-class router. Each class is owned by the model that performs best on held-out evaluation. The legacy five labels (SKILL, CERT, DEGREE, EMPLOYER, YEARS_EXP) go to the DistilBERT v6 ensemble. TOOL goes to ModernBERT v11. The Gate 2 labels (INDUSTRY, LOCATION, PROJECT, SOFT_SKILL) go to the GLiNER 10 type candidate. This is not a single monolithic model dressed up as an ensemble; it is an explicit, hard ownership decision per label class, backed by per-class F1 against an adjudicated dev set.

4. The Agent brain layer

The brain is an orchestrator of over 21 tools. Five end-to-end workflows: joint hire analysis, workforce risk scan, quarterly retention plan, candidate ranking, and candidate-to-multiple-jobs triage. A sixth workflow executes three separate decision procedures simultaneously: a sophisticated 20-cell matrix, a conservative 20-cell matrix, and a naive two-signal heuristic. It then detects disagreements to trigger automatic escalation. When all three procedures agree, it indicates strong consensus. If two procedures agree and one differs, the process escalates to a senior reviewer. When all three procedures disagree, it escalates to a hiring committee. The pattern emerged from my reading of EU AI Act Article 14 and of multiple specialized expert consensus methods, and it adds about 80 milliseconds of latency for a meaningful safety property.

A counterfactual flip explainer fires automatically on DEFER and DECLINE verdicts. It lists the minimal single-lever adjustments needed to shift the (fit tier by team risk) cell into a positive value. Two types of flip levers are evaluated. Candidate side, where a named skill or certification gap is closed, lifting the candidate's tier into a hire family.

Team side, where a retention intervention drops the team's predicted attrition probability across a band threshold. Each path is graded as high, medium, or low feasibility based on its causal status (causal, mixed, or correlational) and the implied lead time. The result is an explainer the hiring committee can act on without asking the model another question.

The Human in the Lead memory layer (`feedback.py`) provides the brain with a small, controlled, traceable, and reversible adaptation channel. Six event types: verdict correction, rule override, confidence calibration, causal update, new intervention, and general comment. Aggregation uses configurable thresholds (auto-apply after three same-direction corrections, confidence recalibration after ten events, causal status flip after five). High-impact events (rule override, new intervention) require explicit approval rather than threshold voting. The brain consults the feedback store before every hiring decision and stamps every memo with a `learning_version`, a `feedback_state_hash`, and a `review_priority`.

5. Evaluation Rigor

The dev set included with this project is not a single snapshot taken at one point in time. Instead, it is a canonical sidecar comprising 25 documents and 840 spans that underwent six pipeline phases: candidate review, repair overlay, quality preflight, tool and skill application, metadata enrichment, and canonical promotion before being stored in `data/processed/`.

The entire process is documented in `docs/EVAL_METHODODOLOGY.md`. Each phase produces its own audit run, including a handoff record, validation report, and provenance log. Once artifacts are finalized, they are locked in read-only mode upon acceptance, and no earlier phase directory has been altered by subsequent phases. The pipeline aligns with the NIST AI Risk Management Framework, emphasizing testing, evaluation, verification, and validation throughout the AI lifecycle, as well as with EU AI Act Article 14 on human oversight of high-risk AI systems. Each span clearly displays its human review status. Out of the 840 retained spans, 31 have an explicit human override record: 22 boundary expansions during repair overlay, one expand and drop conflict pair, two outer survivors of nested span resolution, five TOOL versus SKILL relabels during policy application, and one current keep with explicit blocker clearance during

payroll resolution. The remaining 809 carry a null override record because they passed the multi stage review pipeline default. The pipeline uses model-assisted annotation under human supervision; the human reviewer is the source of truth at every decision point. Production inference itself is fully local and does not depend on any remote model. The integrity check below is recomputable from the canonical sidecars that ship with this submission. Every claimed value matches the actual file on disk.

Check	Claimed	Actual	Match
labels.json sha256	4aa8f79c, 2,841,532 bytes	4aa8f79c, 2,841,532 bytes	yes
provenance.jsonl sha256	6dd14b1f, 2,115,664 bytes (840 lines)	6dd14b1f, 2,115,664 bytes	yes
manifest.json sha256	259fff16, 12,630 bytes	259fff16, 12,630 bytes	yes
eval_dev_v1.json sha256 (input integrity)	0b5a643f, 456,470 bytes	0b5a643f, 456,470 bytes	yes
Record and line counts	840 records, 840 provenance lines, 25 docs	840 / 840 / 25	yes
Per label totals (SKILL 311, TOOL 277, ...)	sums to 840	sums to 840	yes
Override breakdown	31 with explicit override, 809 with null	31 / 809	yes
Round trip masked_text[s:e] equals text	840 of 840 pass	840 of 840 pass	yes
Duplicate spans across documents	0	0	yes

Overlap or nesting in retained set	0 pairs	0 pairs	yes
Per document masked text hash match	25 of 25	25 of 25	yes
Forbidden source key deep walk on every emitted JSON	0 hits	0 hits	yes
Schema version	v7 (10 labels)	v7 (10 labels)	yes
Blind set discipline	never opened, read, stat'd, or hashed	confirmed	yes

The reason this matters for an HR system is the same reason it matters for any high-risk AI deployment. A hire or do-not-advance recommendation is contestable. When a department head, an internal auditor, or a labor relations attorney asks, "How do I know the model is right?" the answer needs to be more than "Trust the model." Every evaluation number cited in the dashboard or in a brain memo traces back to a specific span in this canonical sidecar, with its review history attached. The integrity bracket above is the receipt.

6. Lesson Learned and Surprise

What worked best was making the deep learning models a pure tool layer beneath an orchestrator that is independently swappable. The same DistilBERT, GLiNER, ModernBERT, SBERT, and Bi LSTM stack serves the proprietary brain, the optional CrewAI path, and the Streamlit boardroom dashboard. When I caught the Bi LSTM double sigmoid bug late in the build, the fix was a one-line change to the model file. Nothing in the agent layer needed to know.

What did not work twice. The Bi LSTM was bimodal at 0.00 and 1.00 across all eight departments for almost two weeks. AUC alone did not flag it. Only when I plotted the probability distribution did the saturation become visible. The cause was a redundant

nn.Sigmoid() in the classification head, along with BCEWithLogitsLoss during training, which already applies sigmoid internally. The forward pass involved computing sigmoid(sigmoid(logit)), which caused gradients on confident examples to reach zero. As a result, the model was adjusted by pushing logits toward positive or negative infinity. After the fix, the spread became 0.12 to 0.57, and the Brier score landed at 0.232, near the theoretical floor for a balanced label. The second false start was the early NER fine-tune reporting an F1 of 0.9999 on a synthetic validation split, which was meaningless because the val set used the same templates as training. After I rebuilt the held-out set from real annotated resumes, the honest held-out F1 dropped to 0.74 for DistilBERT alone and 0.78 for the ensemble.

The surprising thing I learned is that "agentic" does not require an LLM. The standard mental model from the course readings is that an agent is an LLM in a loop. Working through the five workflows in brain.py showed me that "agentic" is really three components wired together: tools, a reasoning loop, and memory. An LLM is just one way to implement the reasoning step, not the only option. Using a 20-cell decision matrix with explicit rules is more defensible in an audited domain, executes in 500 milliseconds instead of 30 seconds, costs \$0 per memo rather than \$0.03 to \$0.10, and eliminates hallucinations because the text is templated from verified tool outputs. In multi-agent systems for high-stakes domains, tool depth substitutes for LLM horsepower in a way that is both underappreciated and easier to engineer.

7. What I would build next

Three key extensions, ranked by their significance for a practical ABB pilot. First, swap out the gradient times input driver attribution with SHAP. While gradient saliency provides a first-order approximation, Shapley values are mathematically sound and more defensible, especially if a department head questions the claim that "75 percent of risk comes from comp ratio." Next, incorporate a survival head into the Bi LSTM. While the existing model estimates if someone will leave, the survival head (using Cox proportional hazards or discrete time Weibull) predicts the timing. A "Median time to exit 4.2 months" offers more actionable insight for CHRO planning than just a binary risk score. Third, build a real Workday or SuccessFactors connector to replace the synthesized monthly HRIS data with live reads covering the previous 24 months. The architecture is designed to support that swap.

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